DMSD

Assignment-3

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1. select C.SID:

```
from catalog c, Part p
where c.PSID = P.P(P) and P.colour = 'red' AND
C.SID IN (SELECT C.SID
          FROM CATALOG C, PART P
          where C.PID = P.PID AND P.COLOR = 'green')
(SELECT C.SID
 FROM CATALOG C, PART P
 where C.PID ~~AND~~ = P.PID AND P.COLOR = 'red')
INTERSECT
(SELECT C.SID
 FROM CATALOG C, PART P
 WHERE C.PID = P.PID AND P.COLOR = 'green')
(SELECT SID
 FROM CATALOG NATURAL JOIN PART
 WHERE COLOR = 'red'
INTERSECT
(SELECT SID
 FROM CATALOG NATURAL JOIN PART
```

WHERE COLOR = 'green')

2.

SELECT C.SID
FROM CATALOG C, PART P
WHERE C.PID = P.PID AND (P.COLOR = 'red'
OR P.COLOR = 'green'

SELECT SID
FROM CATALOG NATURAL JOIN PART
WHERE COLOR = 'red' OR P.COLOR = 'green'

~~SELECT SID~~
~~FROM CATALOG NATURAL~~

(SELECT C.SID
FROM CATALOG NATURAL JOIN PART
WHERE C.PID = P.PID AND P.COLOR = 'red')
UNION
(SELECT C.SID
FROM CATALOG C, PART P
WHERE C.PID = P.PID AND P.COLOR = 'green')

3.

```
SELECT S.SID
FROM SUPPLIER S
WHERE NOT EXISTS (SELECT P.PID
                  FROM PARTS P
                  WHERE P.COLOUR='red' OR
                        P.COLOR = 'green')
                  EXCEPT
                  (SELECT C.PID
                  FROM CATALOG C
                  WHERE C.SID = S.SID
```

```
SELECT S.SID
FROM SUPPLIER S
WHERE NOT EXISTS (SELECT *
                  FROM PARTS P
                  WHERE P.COLOR='red' OR
                        P.COLOR = 'green'

                  AND NOT EXISTS (SELECT *
                  FROM CATALOG C
                  WHERE C.SID = S.SID AND
                        C.PID = P.PID))
```

4.

```
SELECT S.SNAME
FROM SUPPLIERS
WHERE EXISTS (SELECT *
              FROM PARTS P
              WHERE P.COLOR='red' AND
              NOT EXISTS (SELECT *
                          FROM CATALOG C
                          WHERE C.SID=S.SID AND
                                C.PID=P.PID))
```

The following view computes the SIDs of suppliers who supply every red part

```
CREATE VIEW SUPPLIERS_ALL_RED_PARTS
SELECT S.SID
FROM SUPPLIERS
WHERE NOT EXISTS ((SELECT P.PID
                   FROM PARTS P
                   WHERE P.COLOR='red')
                   EXCEPT
                   (SELECT C.PID
                   FROM CATALOG C
                   WHERE C.SID=S.SID))
```

(OR)

```
CREATE VIEW SUPPLIERS_ALL_RED_PARTS
SELECT S.SID
FROM SUPPLIER S
WHERE NOT EXISTS (SELECT *
                  FROM PARTS P
                  WHERE P.COLOR = 'red' AND
                  NOT EXISTS (SELECT *
                              FROM CATALOG C
                              WHERE C.SID = S.SID
                              AND C.PID = P.PID))
```

Then the query is computed as follows:

```
SELECT SNAME
FROM SUPPLIER
WHERE SID NOT IN SUPPLIERS_ALL_RED_
                 PARTS.
```

5.

```
SELECT S.SID, S.SNAME, AVG (COST) AS TOTAL
FROM SUPPLIER S, CATALOG C
WHERE S.SID = C.SID AND
C.SID NOT IN (SELECT C1.SID
              FROM CATALOG C1, PARTS P1
              WHERE C1.SID = S1.SID AND
              P1.COLOR <> 'Red')
```

GROUP BY S.SID, S.SNAME

SELECT SID, SNAME, AVG (COST) AS TOTAL
FROM SUPPLIER NATURAL JOIN CATALOG
WHERE SID NOT IN (SELECT SID
FROM CATALOG NATURAL
JOIN PARTS
WHERE COLOR <> 'Red')
GROUP BY SID, SNAME

6. SELECT C.PID, S.SNAME
FROM SUPPLIER S, CATALOG C
WHERE S.SID = C.SID AND C.COST <
(SELECT MAX(COST)
FROM CATALOG
WHERE PID = C.PID

7. SELECT C.PID, S.SID, S.SNAME
FROM SUPPLIERS S, CATALOG C
WHERE S.SID = C.SID AND
C.PID IN (SELECT PID
FROM CATALOG NATURAL
JOIN SUPPLIERS

```
C.COST = (SELECT MAX(COST)
          FROM CATALOG
          WHERE PID = C.PID)
```

8.
```
SELECT PID, PNAME, COUNT(SID)
FROM PARTS NATURAL JOIN CATALOG
GROUP BY PID
HAVING COUNT(SID) >= 2
```

9.
```
(SELECT P.PID
 FROM PARTS P
 WHERE NOT EXISTS (SELECT SID FROM
                   SUPPLIERS WHERE ADDRESS =
                   'Newark'
                  EXCEPT
                  (SELECT SID FROM CATALOG
                   WHERE PID = P.PID)))
UNION
(SELECT P.PID
 FROM PARTS P
 WHERE NOT EXISTS (SELECT SID FROM
                   SUPPLIERS where Address = 'Trenton'
```

```
EXCEPT
(SELECT SID FROM CATALOG
 where PID = P.PID)))

(SELECT P.PID
FROM PARTS P
where NOT EXISTS (SELECT *
                  FROM SUPPLIERS S
                  WHERE S.ADDRESS =
                        'Newark' AND
                  NOT EXISTS (SELECT *
                              FROM CATALOG C
                              WHERE C.PID = P.PID AND
                                    C.SID = S.SID)))

UNION
(SELECT P.PID
FROM PARTS P
WHERE NOT EXISTS (SELECT *
                  FROM SUPPLIERS S
                  where S.Address = 'Trenton'
                        AND
                  NOT EXISTS (SELECT *
```

```
FROM CATALOG C
WHERE C.PID = P.PID AND C.SID =
                S.SID)))
```

10.

```
(SELECT P.PID
FROM PARTS P
WHERE NOT EXISTS (SELECT SID
                  FROM SUPPLIERS
                  WHERE ADDRESS = 'Newark'
                  OR ADDRESS = 'Trenton'
                  EXCEPT
                  (SELECT SID
                  FROM CATALOG
                  WHERE PID = P.PID)))
(SELECT P.PID
FROM PARTS P
WHERE NOT EXISTS (SELECT *
                  FROM SUPPLIERS S
                  WHERE S.ADDRESS = 'Newark'
                  OR S.ADDRESS = 'Trenton' AND
                  NOT EXISTS (SELECT *
                  FROM CATALOG C
```

```
WHERE = C.PID = P.PID AND
        C.SID = S.SID)))

(SELECT P.PID
FROM PARTS P
WHERE NOT EXISTS (SELECT SID FROM
    SUPPLIERS where ADDRESS = 'Newark'
        EXCEPT
        (SELECT SID FROM CATALOG
        WHERE PID = P.PID)))

INTERSECT
FROM PARTS P
WHERE NOT EXISTS (SELECT *
        FROM SUPPLIERS S
        WHERE S.ADDRESS = 'Newark'
    OR S.ADDRESS = 'Trenton' AND
    NOT EXISTS (SELECT *
        FROM CATALOG C
        WHERE C.PID = P.PID AND
        C.SID = S.SID)))
```

```
(SELECT P.PID
FROM PARTS P
WHERE NOT EXISTS (SELECT SID FROM
    SUPPLIERS WHERE ADDRESS = 'Trenton'
    EXCEPT
    (SELECT SID FROM CATALOG WHERE
                PID = P.PID)))
```

11.

```
SELECT SID
FROM PARTS NATURAL JOIN CATALOG
WHERE COLOR = 'red' AND
    SID NOT IN (SELECT SID
        FROM PARTS NATURAL
        JOIN CATALOG
        WHERE COLOR = 'blue')
(SELECT SID
FROM PARTS NATURAL JOIN CATALOG
WHERE COLOR = 'red')
EXCEPT
(SELECT SID
  FROM PARTS NATURAL JOIN CATALOG
  WHERE COLOR = 'blue')
```

12.

```
SELECT S.SID, S.SNAME, C.PID
FROM SUPPLIERS S, CATALOG C
WHERE S.SID = C.SID AND
      (SELECT COUNT (PID)
       FROM CATALOG
       WHERE SID = S.SID) > 4 AND
       C.COST = (SELECT MAX(COST)
                 FROM CATALOG
                 WHERE SID = S.SID)
```

13.

```
SELECT COLOR, COUNT (SID)
FROM PARTS NATURAL JOIN CATALOG
GROUP BY COLOR.
```

14.

```
SELECT SID
FROM PARTS NATURAL JOIN CATALOG
GROUP BY SID
HAVING COUNT (DISTACT P.COLOR) >= 2
SELECT SID
FROM (CATALOG NATURAL JOIN PARTS)
      NATURAL JOIN
```

(CATALOG AS C1(PID, SID1, COST1)

NATURAL JOIN PART AS

P1(PID1, PNAME1, COLOR1)))

WHERE COLOR <> COLOR1

15. SELECT PID, PNAME, MAX(COST),

MIN(COST)

FROM PARTS NATURAL JOIN CATALOG

GROUP BY PID.